

DONGDUCHEON, Korea, Republic ofWMO: **470980**Lat: **37.9019N**Lon: **127.0608E**Elev: **362**StdP: **14.50**Time Zone: **9.00 (E09)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 6.1	(c) 9.9	(d) -10.3	(e) 3.2	(f) 15.6	(g) -6.5	(h) 3.9	(i) 17.6	(j) 14.3	(k) 25.8	(l) 13.0	(m) 26.5	(n) 2.8	(o) 20	(p) 0.384

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 13.8	(c) 89.8	(d) 75.4	(e) 87.3	(f) 73.5	(g) 85.0	(h) 72.0	(i) 78.2	(j) 85.6	(k) 76.8	(l) 83.8	(m) 75.6	(n) 82.1	(o) 4.8	(p) 290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.2	138.8	82.1	74.8	132.2	80.7	73.6	126.7	79.8	42.1	85.5	40.6	84.1	39.5	82.2	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 13.7	(b) 11.7	(c) 10.0	(e) 1.6	(f) 93.7	(g) 4.7	(h) 2.7	(i) -1.7	(j) 95.6	(k) -4.5	(l) 97.1	(m) -7.1	(n) 98.6	(o) -10.5	(p) 100.6		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
					0.3	80.2	4.3	1.5	-2.8	81.3	-5.3	82.2	-7.6	83.0	-10.7	84.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.3	25.7	31.5	41.7	53.0	63.6	71.5	76.1	77.4	68.8	57.0	43.1	29.2
	DBStd	18.99	7.66	7.44	7.07	6.40	4.95	3.81	3.94	4.44	4.58	6.26	7.98	7.93
	HDD50	2480	752	519	272	40	0	0	0	0	0	16	235	646
	HDD65	5369	1217	938	723	362	85	3	0	0	18	253	659	1111
	CDD50	3700	0	1	14	131	423	646	811	849	564	235	26	0
	CDD65	1113	0	0	0	2	43	199	346	384	132	7	0	0
	CDH74	8264	0	0	0	46	459	1387	2422	3190	722	38	0	0
	CDH80	2270	0	0	0	4	78	347	677	1062	101	1	0	0
(14) Wind	WSAvg	3.5	3.2	3.8	4.6	4.5	4.0	3.5	3.4	3.3	2.8	2.8	3.3	3.2
(15) Precipitation	PrecAvg	58.3	0.7	1.1	1.6	3.8	3.5	5.9	17.8	13.3	6.1	2.0	1.7	0.9
	PrecMax	84.7	2.0	2.8	3.8	6.7	6.0	15.2	43.4	35.5	19.3	6.0	4.4	2.7
	PrecMin	29.4	0.1	0.0	0.5	1.2	0.3	1.6	6.2	1.5	1.1	0.6	0.3	0.2
	PrecStd	13.7	0.6	0.8	0.9	1.5	1.6	3.2	9.3	9.3	4.6	1.3	1.1	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.3	57.1	67.9	79.1	85.8	89.0	92.2	93.9	85.3	77.8	67.7	52.6
		MCWB	42.0	45.8	50.0	58.0	62.9	67.6	76.9	77.9	69.6	62.4	56.1	44.6
	2%	DB	43.4	51.1	62.3	74.0	82.1	86.2	88.9	91.2	82.7	74.4	62.7	47.8
		MCWB	36.3	40.4	48.9	55.4	61.8	67.4	75.8	77.2	68.9	61.1	52.6	41.5
	5%	DB	40.1	47.1	58.1	69.7	79.1	84.0	86.2	88.5	80.7	71.2	59.0	44.3
		MCWB	34.0	38.5	46.2	52.6	60.5	67.1	74.7	76.0	67.6	59.2	50.8	38.5
	10%	DB	37.3	43.1	54.2	66.2	76.0	81.7	83.8	85.7	78.5	68.3	56.0	41.2
		MCWB	31.8	35.7	43.4	51.0	59.5	66.6	73.7	74.7	66.2	57.5	49.1	35.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.1	50.6	54.6	61.8	67.3	73.1	79.5	80.9	75.3	67.2	59.1	46.4
		MCDB	47.2	54.3	61.8	72.9	76.4	81.1	87.0	90.0	80.3	71.1	64.3	50.5
	2%	WB	37.8	42.1	50.9	58.3	65.0	71.3	77.6	79.0	72.2	63.2	55.4	42.5
		MCDB	42.2	49.1	59.4	68.6	75.3	79.5	84.3	87.2	77.6	70.2	60.4	47.0
	5%	WB	35.0	38.9	47.7	55.6	63.2	70.0	76.4	77.6	70.3	61.2	52.4	39.1
		MCDB	38.8	45.9	55.7	64.5	74.0	78.8	83.0	85.2	76.4	68.7	58.0	43.8
	10%	WB	32.3	36.4	44.8	53.2	61.5	68.8	75.4	76.4	68.9	59.1	49.7	36.4
		MCDB	36.5	42.3	52.5	63.2	72.1	77.5	81.7	83.3	75.3	66.7	54.7	40.8
(35) Mean Daily Temperature Range	MDBR		18.0	18.8	20.1	21.4	21.4	17.9	12.0	13.8	17.6	20.9	18.1	16.8
	5% DB	MCDBR	19.0	22.9	25.3	27.7	27.5	23.2	17.0	17.4	20.5	23.1	21.1	18.4
		MCWBR	12.8	14.0	13.4	12.5	10.9	8.5	6.6	6.3	8.5	10.8	12.6	13.1
	5% WB	MCDBR	15.9	20.4	20.9	22.4	20.7	16.4	12.7	14.1	14.7	19.4	17.6	17.2
		MCWBR	11.7	14.0	12.8	12.3	9.9	7.1	5.8	6.1	7.4	10.3	12.3	13.3
(40) Clear-Sky Solar Irradiance	taub		0.426	0.487	0.571	0.622	0.631	0.703	0.674	0.580	0.506	0.502	0.479	0.425
	taud		1.921	1.823	1.696	1.635	1.661	1.584	1.692	1.907	2.027	1.918	1.913	1.957
	Ebn at Noon		222	225	219	218	220	204	209	226	236	218	203	212
	Edh at Noon		45	56	70	79	78	85	75	60	50	50	45	41
(44) All-Sky Solar Radiation	RadAvg		786	1034	1337	1562	1756	1641	1269	1352	1302	1124	765	692
	RadStd		51	77	131	117	167	163	190	186	143	79	78	52

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+1.35	N/A	N/A	+2.00	+1.58	N/A	N/A	N/A	+385	+243
(47) Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+191	+104

Nomenclature: See separate page